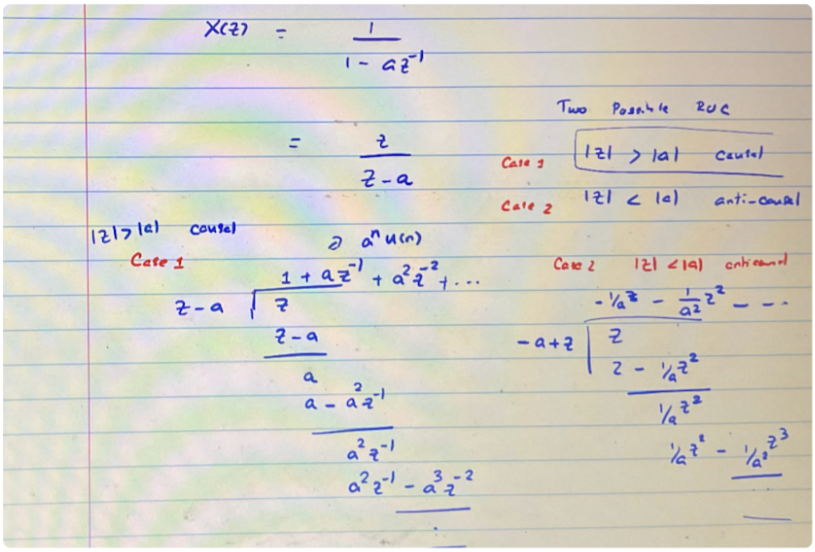
- $X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$ DTFT $\omega \to$ real, $X(\omega) \to$ complex
- $x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$ • $X(\omega)$ has period $= 2\pi$
- $X(e^{j\omega}) = x(\omega)$
  ↓
  z-transform @ $z = e^{j\omega}$ and $|z| = 1$ in ROC
- $x(n) \to$ absolutely summable, $\sum_{n=-\infty}^{\infty} |x(n)|^2 < \infty \to$ DTFT exists
- $x(n) = a^n u(n) \Rightarrow X(\omega) = \frac{1}{1 - ae^{j\omega}} = \frac{1}{1 - a\cos\omega + ja\sin\omega}$
- $x(n) = e^{j\omega_0 n} \Rightarrow X(\omega) = \sum_{k=-\infty}^{\infty} 2\pi \delta(\omega - (\omega_0 + k \cdot 2\pi))$
- $x_1(n) \leftrightarrow X_1(\omega)$, $x_2(n) \leftrightarrow X_2(\omega) \to (ax_1 + bx_2)(n) \leftrightarrow aX_1(\omega) + bX_2(\omega)$
- $x_1(n) = x(n - n_0) \rightleftarrows X_1(\omega) = e^{-j\omega n_0} X(\omega)$
- $x_1(n) = x(-n) \Rightarrow X_1(\omega) = X(-\omega)$ • $x_c(n) \leftrightarrow x_R(\omega)$
- $x_1(n) = x^*(n) \Rightarrow X_1(\omega) = X^*(-\omega)$ • $x_a(n) \leftrightarrow jX_I(\omega)$
- $x(n)$ is real → $|X(\omega)| \to$ even ; $\angle X(\omega) \to$ odd
- $x(n)$ is real & even $\Rightarrow X(\omega) \to$ real, even
- $x(n)$ is real & odd $\Rightarrow X(\omega) \to$ img, odd
- $x(n) e^{j\omega_0 n} \Leftrightarrow X(\omega - \omega_0)$ • $nx(n) \leftrightarrow j \frac{d}{d\omega} X(\omega)$
- $x(n) * h(n) \longleftrightarrow X(\omega) H(\omega)$
- $x(n) \cdot h(n) \longleftrightarrow \frac{1}{2\pi} \int_{\theta=-\pi}^{\pi} X(\theta) H(\omega - \theta) d\theta$ (periodic convolution)
- $\sum_{n=-\infty}^{\infty} x(n) y^*(n) = \frac{1}{2\pi} \int_{\omega=-\pi}^{\pi} X(\omega) Y(\omega) d\omega$ • $\sum_n |x(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\omega)|^2 d\omega$

- Ideal LPF: $H(\omega) = \begin{cases} 1 ; & |\omega| \le \omega_0 \\ 0 ; & \omega_0 < |\omega| \le \pi \end{cases} \Rightarrow h(n) = \frac{\sin(\omega_0 n)}{\pi n}$ sinc fun^n (unstable)

- window: $w(n) = \begin{cases} 1, & |n| \le M \\ 0 ; & \text{else} \end{cases}$ $W(\omega) = \frac{\sin\left(\frac{\omega}{2}(2M+1)\right)}{\sin(\omega/2)}$
  signal

- Ideal HPF: $H_{HP}(\omega) = H_{LP}(\omega - \pi) \Rightarrow h_{hp}(n) = h_{LP}(n)(-1)^n$

- $X(z) = \frac{P(z)}{Q(z)} = \frac{a_0 z^N + a_1 z^{N-1} + \cdots}{b_0 z^M + b_1 z^{M-1} + \cdots}$ (Roc R) let $b_0 = 1$, $a_0 \neq 0$
  use partial fractions

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  long division method
  - contour integration $x(n) = \frac{1}{2\pi} \int_{\omega=0}^{2\pi} x(re^{j\omega}) r^n e^{j\omega n} d\omega$
  - power series method

- Initial val th: $x(n)$ is causal $\Rightarrow \lim_{z \to \infty} x(z) = x(0)$
- Final val th: $x(n)$ is causal $\Rightarrow x(\infty) = \lim_{N \to \infty} x(N)$
- Inverse LTI: $h(n) * g(n) = \delta(n) \Rightarrow G(z) = 1/H(z)$
